

# Shikhar Veeramachineni

+91 9381313981 — shikharveer01@gmail.com — [linkedin.com/in/shikhar-veeramachineni](https://www.linkedin.com/in/shikhar-veeramachineni) — [github.com/ShikharVeer10](https://github.com/ShikharVeer10) —  
shikhar-veerdeve.vercel.app —  
Hyderabad, India

## PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with experience developing agentic AI systems, document intelligence platforms, retrieval-augmented applications, and large language model workflows. Skilled in building scalable backend architectures using FastAPI, PostgreSQL, REST APIs, and asynchronous processing while integrating modern AI technologies including PydanticAI, OpenAI, Gemini, Transformer models, and multi-agent orchestration frameworks. Strong interest in AI infrastructure, backend engineering, autonomous systems, NLP, and production-grade machine learning applications.

## EXPERIENCE

### AI Intern

May 2026 – July 2026

*Deloitte*

*Hyderabad*

- Designed and developed a document extraction pipeline capable of processing PowerPoint presentations and extracting structured information including text, headings, bullet hierarchies, tables, shapes, coordinates, and formatting metadata.
- Built normalization and layout reconstruction modules to convert heterogeneous document elements into a unified JSON schema, preserving spatial relationships, grouping structures, and document hierarchy.
- Implemented visual and structural analysis techniques to capture style attributes such as font properties, colors, backgrounds, section layouts, and table structures for enterprise-grade document understanding.
- Developed relationship detection logic to identify process flows, mappings, parent-child hierarchies, and connections between document elements, enabling representation of complex business workflows in machine-readable form.
- Integrated LLM-based summarization and classification capabilities to generate slide-level and document-level insights, supporting downstream applications such as knowledge retrieval, analytics, and intelligent document search.

### Summer Trainee

June 2025 – July 2025

*Grant Thornton Bharat LLP*

*Hyderabad, India*

- Worked on analyzing internal business datasets and supporting recurring reporting workflows by developing Python-based visualizations, automated data processing scripts, and structured ETL pipelines. Collaborated with consulting teams to translate business requirements into technical analytics solutions, dashboards, and actionable reporting frameworks.
- Worked closely with consultants on the PMAY (Pradhan Mantri Awas Yojana) project, analyzing large-scale datasets related to housing allocation, beneficiary records, project milestones, and progress tracking. Developed interactive visualizations and analytical reports that supported operational reviews, performance monitoring, and stakeholder decision-making.
- Designed and implemented normalized SQL database schemas to enable efficient storage, querying, aggregation, and retrieval of PMAY-related data. Optimized data models and query structures to improve reporting performance, data consistency, and cross-functional accessibility.
- Developed dashboard-ready datasets and DPR tracking mechanisms by integrating data from multiple operational sources, enabling consultants and project teams to monitor program performance, beneficiary coverage, allocation trends, and execution progress in a standardized manner.
- Created Matplotlib-based visual analytics and custom reporting components to identify trends, anomalies, and performance indicators within large-scale government program datasets, helping teams derive actionable insights from complex data.

### AI Intern

Apr 2023 – June 2023

*Coschool*

*Hyderabad, India*

- Designed and optimized prompt engineering workflows for Generative AI models to create curriculum-aligned educational content, including question generation, concept explanations, quizzes, and learning materials across multiple academic subjects.
- Developed structured prompt templates and evaluation frameworks to improve the quality, consistency, and pedagogical relevance of AI-generated outputs, ensuring alignment with educational objectives and learning outcomes.
- Evaluated and benchmarked AI-generated responses by analyzing factors such as correctness, coherence, difficulty level, curriculum coverage, and instructional effectiveness, providing recommendations for prompt improvements.
- Collaborated with AI and content development teams to identify failure cases, reduce hallucinations, and improve the reliability of generated educational materials through systematic prompt tuning and output validation.
- Assisted in developing training materials and demonstrations showcasing the effective use of Generative AI tools such as ChatGPT for educational content creation, lesson planning, and learner engagement.

## PROJECTS

- Emotio** | Python, Pandas, NumPy, NLTK, Tweepy (Twitter API), Hugging Face Transformers, CardiffNLP RoBERTa, KeyBERT, Scikit-learn. 2026
- Built Emotio, an AI-powered sentiment analysis platform that classifies user opinions into positive, neutral, and negative sentiments.
  - Leveraged Transformer-based NLP models to perform accurate sentiment prediction on real-world social media text.
  - Developed an end-to-end pipeline for data collection, preprocessing, inference, and sentiment visualization.
  - Implemented keyword and topic extraction to identify the main themes driving user sentiment.
  - Designed an interactive dashboard to provide real-time sentiment insights and trend analysis for decision-making.
- DeepRoute** | Python, FastAPI, Machine Learning, Route Optimization, Geospatial Processing, REST APIs 2026
- Built an intelligent route optimization platform that analyzes travel paths and generates efficient routing recommendations using data-driven decision-making techniques.
  - Implemented geospatial data processing workflows to evaluate route efficiency, travel distance, and path selection across multiple routing scenarios.
  - Integrated machine learning and optimization logic to improve route recommendations by considering travel constraints, location relationships, and routing performance metrics.
- Traction AI** | Next.js, FastAPI, PostgreSQL, SQLAlchemy, Alembic, OpenAI, Pydantic-AI, Jinja2 2026
- Built an AI-powered platform that transforms raw startup ideas into structured business assets, including business models, investor summaries, customer personas, problem statements, positioning strategies, and pitch narratives.
  - Engineered a multi-stage backend orchestration pipeline to generate, validate, persist, and manage business artifacts while supporting shareable workspaces and dynamically generated pitch pages.
  - Leveraged validation layers and schema-first prompt engineering techniques to produce context-aware outputs grounded in founder inputs, market assumptions, business constraints, and venture readiness indicators.
- AetherMind** | Python, FastAPI, OpenAI API, Multi-Agent Systems, Prompt Engineering, Workflow Automation, AsyncIO, REST APIs, 2026
- Built an enterprise document extraction system that processes PowerPoint presentations and converts slide content into structured JSON representations for downstream analytics, search, and AI workflows.
  - Developed extraction pipelines to capture slide text, tables, images, positional coordinates, styling metadata, and document structure while preserving layout and visual relationships between elements.
  - Integrated AI-powered slide summarization using PydanticAI and Gemini models, enabling automated generation of contextual slide-level insights and document understanding.
  - Designed validation layers and schema-first prompt engineering techniques to produce context-aware outputs grounded in slide content, document structure, and extraction parameters.
- SynapseArena** | Python, PydanticAI, Pydantic, OpenAI, Gemini 2026
- Architected a multi-agent AI reasoning platform using PydanticAI, enabling specialized autonomous agents to collaboratively debate, critique, and validate complex topics through structured interactions.
  - Engineered an agent orchestration pipeline consisting of Researcher, Optimist, and Skeptic agents that generate evidence-backed arguments, counterarguments, and reasoning artifacts for informed decision-making.
  - Implemented schema-driven AI workflows using Pydantic models, validators, enums, and nested schemas to enforce type-safe communication and reliable structured outputs across agents.
- AutoStream Agent** | Python, Gemini API, LangChain, FAISS, Sentence Transformers, RAG, Conversational AI 2026
- Built a conversational AI sales agent that combines Retrieval-Augmented Generation (RAG), intent detection, and lead qualification workflows to guide users from product inquiries to conversion-ready interactions.
  - Engineered a knowledge-grounded retrieval pipeline using FAISS vector search and Sentence Transformers embeddings, enabling accurate and context-aware responses based on product documentation.
  - Implemented automated lead qualification workflows that collect user information, identify high-intent prospects, and trigger backend tool execution for CRM-style lead capture processes.
- Intellipost** | Python, FastAPI, LiveKit, Deepgram, Cartesia, GPT Agents, WebSocket 2025
- Built an AI-powered document intelligence platform that enables users to upload policy and enterprise documents, extract structured information, and interact with content through API-driven workflows.
  - Engineered document processing pipelines for PDF ingestion, text extraction, metadata generation, and structured content representation to support intelligent document navigation and analysis.
  - Implemented structured document understanding mechanisms that transform unstructured document content into machine-readable representations suitable for search, analytics, and downstream AI workflows.

PUBLICATIONS

Real-Time Twitter Sentiment Analysis with a Hybrid Transformer Framework2026INDIACom-2026

Author: Veeramachineni Shikhar

- Proposed a hybrid Transformer-based sentiment analysis framework for real-time classification of Twitter data using contextual language representations and NLP preprocessing pipelines.
- Designed an end-to-end workflow for tweet collection, text preprocessing, sentiment prediction, keyword extraction, and real-time sentiment monitoring.
- Leveraged Transformer architectures to improve sentiment classification accuracy and contextual understanding across diverse social media conversations.
- Explored scalable approaches for social media analytics, opinion mining, and trend identification using AI-driven language modeling techniques.

EDUCATION

Vellore Institute of Technology – AP2023 – 2027

B.Tech in Computer Science and EngineeringAmaravati

Oakridge International School2011 – 2023

High School DiplomaHyderabad

RECOGNITION

Post-a-thon | 3rd Place2025

- Secured 3rd place in the Post-a-thon hackathon by building IntelliPost, recognized for real-world applicability, system design, and effective use of AI-driven document intelligence.

TECHNICAL SKILLS

**Languages:** Python, SQL, C++, Java

**AI/ML & Agentic Systems:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, RoBERTa, KeyBERT, PydanticAI, Prompt Engineering, RAG, Multi-Agent Systems, Agent Orchestration, Structured LLM Outputs, Autonomous AI Workflows, Embeddings, Vector Search

**Backend & APIs:** FastAPI, REST APIs, AsyncIO, Pydantic, SQLAlchemy, JWT Authentication

**Databases & Vector Stores:** PostgreSQL, SQLite, ChromaDB, FAISS

**Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn

**Computer Vision & Document Intelligence:** OpenCV, YOLO, OCR, PPTX Processing, PDF Processing, Document Extraction, Structured Data Extraction

**LLMs & AI Platforms:** OpenAI API, Gemini API, Ollama, Hugging Face, Transformer Models

**Tools & DevOps:** Git, GitHub, Docker, Postman, VS Code